THE UNIVERSITY OF BRITISH COLUMBIA
Faculty of Forestry & Environmental Stewardship

Dear Dr. Öpik,

Please consider our paper, "Warmer and longer growing seasons increase growth for juvenile trees in a common garden" for publication as a 'Full paper' in *New Phytologist*.

Anthropogenic climate change has lengthened growing season of most temperate trees, which should generally increase growth (Gao *et al.*, 2022). Recent evidence, however, has showed that trees may not grow more with warmer seasons (Dow *et al.*, 2022; Silvestro *et al.*, 2023; Matula *et al.*, 2023), highlighting gaps in our basic model of how season length affects tree growth. Alongside this, other new work suggests earlier seasons may not always be longer growing seasons (Zohner *et al.*, 2023). Here we use a rare combination of full growing season length data via vegetative phenology, and growth via tree rings to address the question how both the start, end, and thermal conditions during the growing season affects growth.

What hypotheses or questions does this work address? Do juvenile trees grow more with longer and warmer seasons? We address this question in a common garden where drought and competition are limited. We test how longer warmer seasons affect growth through shifts in the start and end of season, and determine whether provenance latitude affects our conclusions.

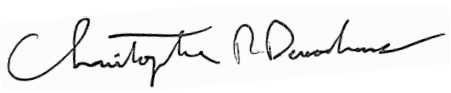
How does this work advance our current understanding of plant science? By a using multi-year, multi-species high quality phenology and growth dataset, we are able to tease apart the contribution of start and end of season events alongside longer versus warmer seasons. Surprisingly, we show that end of season events—most often overlooked—appeared to be more important for growth than spring events by lengthening the thermal season.

Why is this work important and timely? Our work brings new insights on how temperature and end of season events may determine whether trees grow more with climate change (Dow *et al.*, 2022; Silvestro *et al.*, 2023; Matula *et al.*, 2023). We show that the end of season appears more related to growth than the start of season, extending new work suggesting both early and late season events together dynamically determine growth (Zohner *et al.*, 2023).

All authors contributed to this work and approved this version for submission. The manuscript is 4228 words with a 201-word abstract, and 4 figures and is not under consideration elsewhere. We hope you find it suitable for publication in *New Phytologist*, and look forward to hearing from you.

Potential reviewers:
Caroline Polgar—carolinepolgar@gmail.com
Amanda Gallinat—agallina@colby.edu
Libby Ellwood—eellwood@floridamuseum.ufl.edu
Loïc D'Orangeville—loic.dorangeville@sbf.ulaval.ca
Julia Green—juliakgreen@arizona.edu

Sincerely,



Christophe Rouleau-Desrochers
MSc in Forestry
Department of Forest & Conservation Sciences
The University of British Columbia

References

Dow, C., Kim, A.Y., D'Orangeville, L., Gonzalez-Akre, E.B., Helcoski, R., Herrmann, V., Harley, G.L., Maxwell, J.T., McGregor, I.R., McShea, W.J., McMahon, S.M., Pederson, N., Tepley, A.J. & Anderson-Teixeira, K.J. (2022). Warm springs alter timing but not total growth of temperate deciduous trees. *Nature*, 608, 552–557.

Gao, S., Liang, E., Liu, R., Babst, F., Camarero, J.J., Fu, Y.H., Piao, S., Rossi, S., Shen, M., Wang, T. & Peñuelas, J. (2022). An earlier start of the thermal growing season enhances tree growth in cold humid areas but not in dry areas. *Nature Ecology & Evolution*, 6, 397–404.

Matula, R., Knířová, S., Vítámvás, J., Šrámek, M., Knít, T., Ulbrichová, I., Svoboda, M. & Plichta, R. (2023). Shifts in intra-annual growth dynamics drive a decline in productivity of temperate trees in Central European forest under warmer climate. *Science of The Total Environment*, 905, 166906.

Silvestro, R., Zeng, Q., Buttò, V., Sylvain, J.D., Drolet, G., Mencuccini, M., Thiffault, N., Yuan, S. & Rossi, S. (2023). A longer wood growing season does not lead to higher carbon sequestration. *Scientific Reports*, 13, 4059.

Zohner, C.M., Mirzagholi, L., Renner, S.S., Mo, L., Rebindaine, D., Bucher, R., Palouš, D., Vitasse, Y., Fu, Y.H., Stocker, B.D. & Crowther, T.W. (2023). Effect of climate warming on the timing of autumn leaf senescence reverses after the summer solstice. *Science*, 381.